



JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination May-2023

Name of School: School of Engineering and Technology

Name of Programme/Course: B.Tech. (Biomedical) / B.Tech. (Biotech) /
B.Tech. (Chemical) / B.Tech. (Civil) / B.Tech. (Mechanical) / B.Tech. (CSE) /
B.Tech. (CSE-AIML) / B.Tech. (CSE-Cyber Security) / B.Tech. (Food Tech.)
/ B.Tech. (Electrical)

Semester: II

Paper Code: BTEEMCO202T22

Title of Paper: Engineering Mechanics

Time: 3 hrs.

Max. Marks: 60

INSTRUCTIONS/NOTES FOR THE EXAMINEE

1. Attempt any five Questions from section A and attempt any four Questions from section B.
2. Use of scientific calculator (Non programmable) is allowed.

Question
Number

SECTION-A

Marks
Allotted

~~Q.1~~ Explain Lami's Theorem.

— (4)

4

~~Q.2~~ Explain Parallelogram law of forces.

— (4)

4

Q.3 Write the principle of virtual work and prove $\delta W = 0$.

4

~~Q.4~~ Define friction Angle and Angle of Repose.

— (4)

4

~~Q.5~~ Explain Newton's first, second and third law of motion.

— (4)

4

Q.6 Discuss about Time of Flight, Max. Height and Range in Projectile Motion.

4

~~Q.7~~ Explain Work, Power and Energy.

— (1)

4

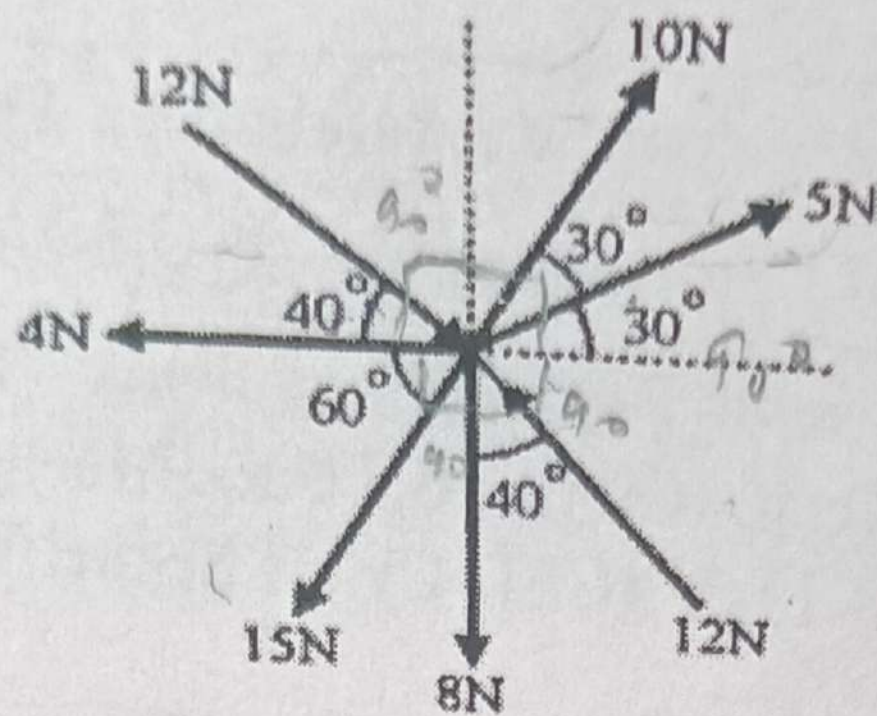
SECTION-B

~~Q.1~~ Classify the various kinds of system of forces with illustrative sketches.

— (5)

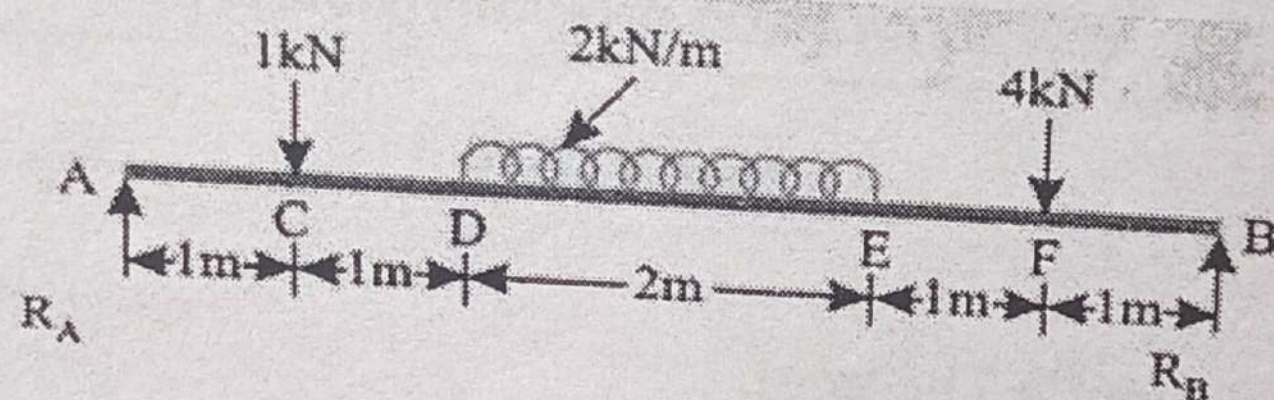
10

- Q.2 Determine the resultant of the force system shown in the figure.

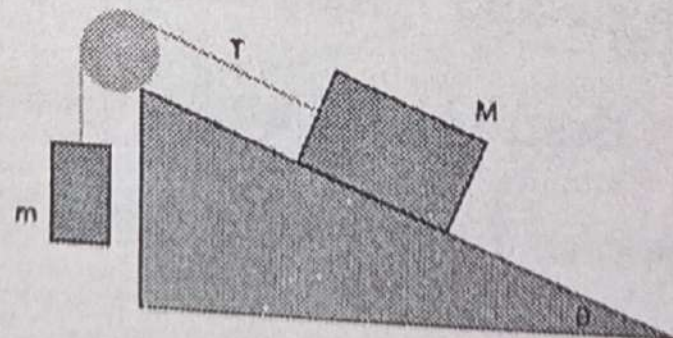


- Q.3 A uniform ladder of weight 800N and of length 7m rests on a horizontal ground and smooth vertical wall. The angle made by the ladder with the horizontal is 60° . When a man of weight 600N stands on the ladder at a distance 4m from the top of the ladder, the ladder is at the point of sliding. Determine the friction between the ladder and the floor.

- Q.4 Find out reaction R_A and R_B by principle of virtual work.



- Q.5 Mass $(M)=50\text{N}$, Mass $(m)=20\text{N}$, inclination angle $(\theta)=30^\circ$, coefficient of friction $(\mu)=0.30$ blocks in equilibrium with frictionless pulley. Determine the friction force.



- Q.6 Prove that the tension ratio in belt friction $(T_1/T_2 = e^{\mu\theta})$.